

BLACK TEA MOTORBIKES

BONFIRE E

User manual version 1.0

Hello & Welcome!

You are now the proud owner of a Black Tea Bonfire E.

Welcome to our rider community. We are very flattered that you have chosen Black Tea Motorbikes for your upcoming adventures. It is now our duty to do our best to provide you with product information, service and spare parts, and offer you a hassle-free experience.

Our mission is to make adventure accessible to everyone, everywhere. That's why our goal is to create compelling motorcycles for any kind of adventure you have in mind. For easy service and personalization, we build our bikes as simple as possible and allow any kind of (legal) modification.

This manual is intended to be your resource and guide for general maintenance, safety precautions, service issues - from what you can do yourself to what you need to leave to professional mechanics - tuning, warranty issues, handling and troubleshooting.

We ask that you use this document to ensure that every ride remains a safe ride.

Black Tea Motorbikes

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1. Introduction

1.1. Safety regulations

We want you and the people around you to get through everyday life safely. That's why we ask you to keep a few things in mind when you're out and about with your Bonfire.

Follow the rules of the road. To minimize the risk of serious injury, always wear protective equipment, including a helmet. Always carry the following documents with you when riding your Bonfire: ID card, insurance papers, license plate and operating permit.

You're probably really looking forward to riding your bike, but please follow the steps below to make sure you don't cause damage to the vehicle or injure others. During shipping, the bike may have been damaged. We cannot guarantee the bike's roadworthiness as it may have been strapped down for up to several weeks before you received it.

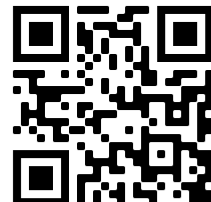
- Read and understand this manual
- Go through the chapter "Regular inspection and maintenance".
- Go through the "Before each trip" checklist.

Make sure you have the latest version of the manual by visiting www.blackteamotorbikes.com/service.

1.2. Unboxing

If you receive a fully assembled motorcycle, you will only need to install the mirrors and reflectors before your first ride. If you receive your Bonfire in a box on a pallet, you will have to unpack it and do the final assembly yourself.

We encourage everyone to watch our how-to videos to learn how to assemble, ride and maintain your bike. Visit our site www.blackteamotorbikes.com/bonfire or scan the QR code here to learn how to use the Bonfire. Remove the cardboard and assemble your Bonfire.



2. General information

2.1. Security & Warranty

Warranty

The standard warranty is covered by the attached Terms and Conditions. It expires in the event of damage caused by actions that contradict these operating instructions.

Usage

Black Tea Motorbikes motorcycles have been homologated to the highest standards for road use by TÜV Süd in Munich. It is therefore prohibited to remove any equipment or tamper with the wiring outside of an authorized service center. The Bonfire is not designed for off-road use or racing.

Important: Although the Bonfire has a fuel cap, the tank is not intended to be filled with any type of liquid. In particular, please do not fill the tank with gasoline. The Bonfire is an all-electric motorcycle.

Driving

When you ride the Bonfire for the first time, there are a few things to keep in mind. The bike has an electric motor with instant torque and a primary drive without gears, which means that you have all the power available at all times through the throttle. It is very sensitive and it is highly recommended to ride in the first driving mode to get used to this type of vehicle.

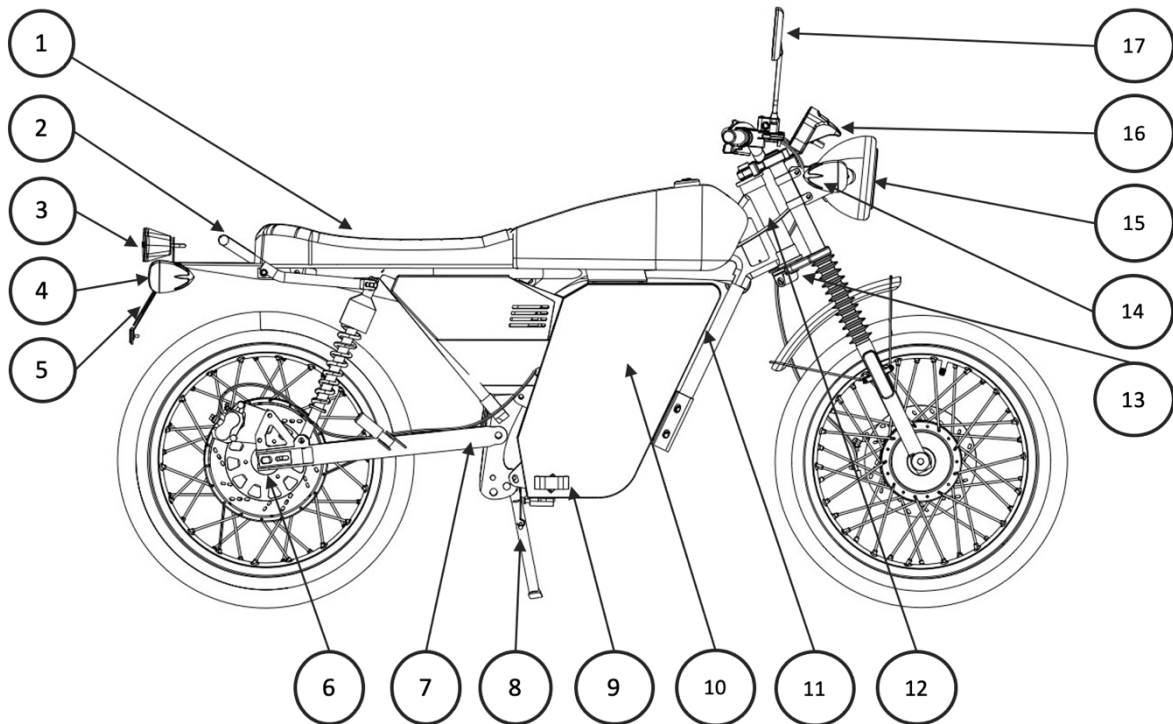
Important: You have two keys. The larger key is for the ignition and steering wheel lock. The small key is for the battery box. Be sure to use the correct key, otherwise you can damage the lock or the key.

There is no clutch; instead, the rear brake is located on the left side of the handlebar. This can feel very unnatural for experienced combustion motorcyclists.

Remember that electric vehicles are quiet. Be aware of people or other cyclists who may be near the lanes you are using.

- Never drive under the influence of alcohol or drugs
- Always check local laws and regulations where you plan to ride

2.2. Motorcycle-Overview



- | | |
|-----------------------------|-------------------------------|
| 1. Seat | 10. Battery box |
| 2. Bracket for passenger | 11. Position of the nameplate |
| 3. Tail and brake light | 12. FIN position |
| 4. Rear turn signals | 13. Steering wheel lock |
| 5. License plate holder | 14. Front turn signals |
| 6. Electric hub motor | 15. Headlight |
| 7. EPA label position (USA) | 16. Display |
| 8. Side stand | 17. Mirror |
| 9. Footrests | |

Vehicle identification number (VIN)

The VIN/chassis number is a 17-digit number stamped on the head tube of the frame. Do not change or remove this number as it is the legal identifier for your motorcycle. You can also find the VIN on the legally required nameplate.

Battery serial number

The battery serial number is located on the battery. The serial number is not visible when the battery is in the motorcycle.

Engine serial number

The serial number of the motor is laser engraved on the left side of the motor housing and is visible.

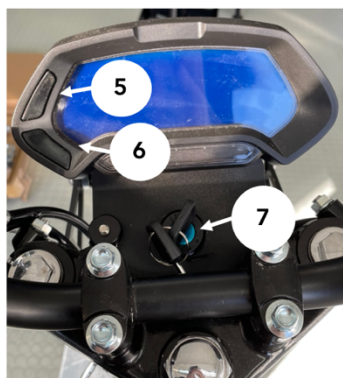
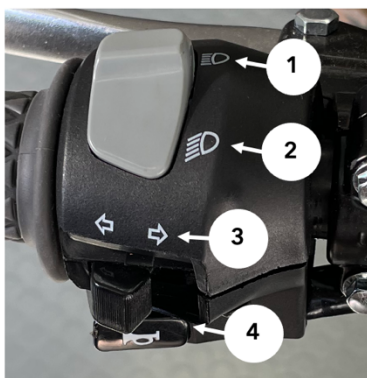
Serial number of the controller

The serial number of the control unit is located on the top of the housing and becomes visible when one of the side covers is removed.

Nameplate

The European and US nameplates are located on the right side of the down tube. The stickers must not be altered or removed.

2.3. Overview of the controls on the handlebar



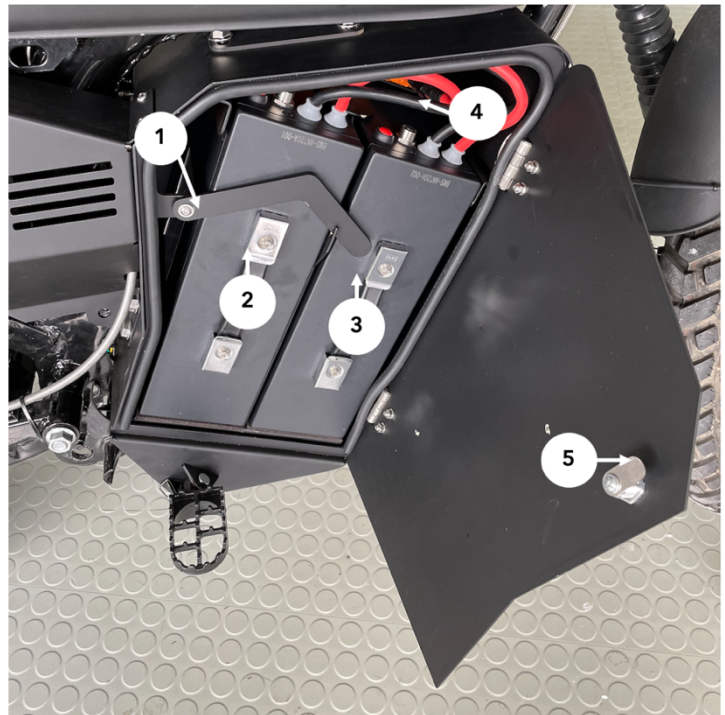
1. High beam
2. Low beam
3. Turn signal
4. Horn

5. setting key
6. selection key
7. ignition lock & Engine control lamp

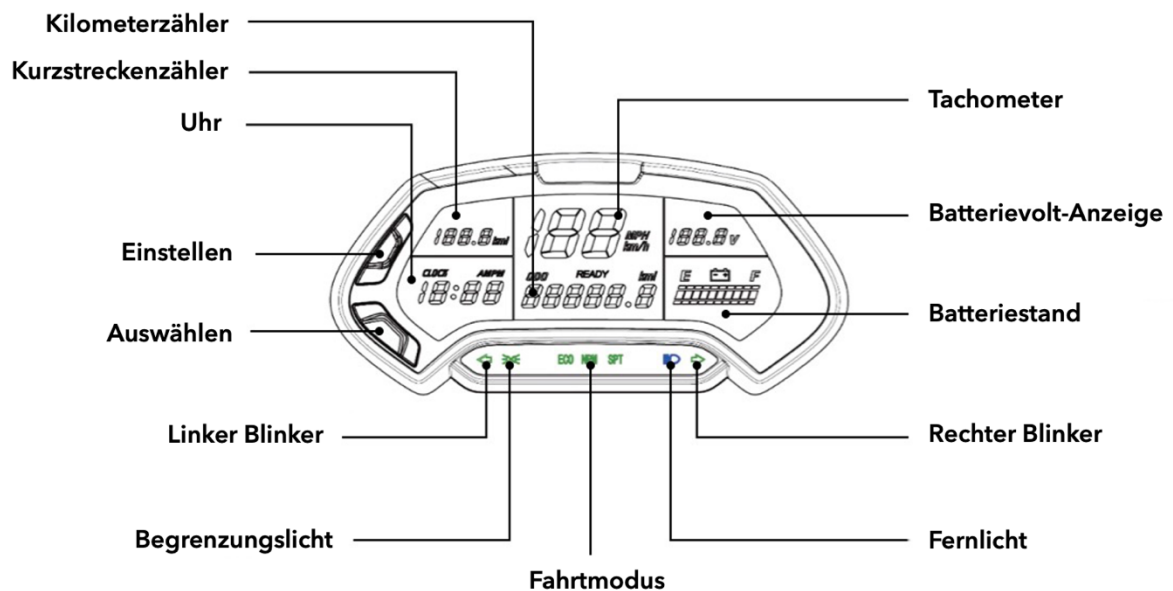
8. kill switch
9. starter
10. driving mode switch

2.4. Battery box overview

1. Battery fuse
2. First battery compartment
3. Second battery compartment
4. Connection to the vehicle
5. Battery box lock



2.5. Display overview



2.6. Vehicle operation

Start your Bonfire

Before you start the motorcycle, please remember that it is an all-electric vehicle that is completely silent when powered on. To start your Bonfire, follow these steps:

1. Make sure that the battery is charged. Place it sideways in the battery box so that the cables are facing up.
2. Move the battery fuse down and close the battery box. If you do not move the battery fuse down, you will not be able to close the box.
3. To close the battery box, you need to push the door slightly against the seal, then you can close the lock by turning it to the left.
4. Make sure that the steering wheel lock and all other locks used are unlocked.
5. Insert the key into the ignition lock and turn it to the right. Now the display and the lights should be on.
6. Fold up the side stand and sit on your bike.
7. Make sure the kill switch is turned on and press the starter.
8. If your riding mode is shown on the display, the bike is now active and you can start riding. If you still cannot start your bike, you may need to raise the side stand.
9. To prevent injuries, there is an additional safety feature. When you stop after riding, "Active" is shown on the display to remind you that the bike is still on.

Important: It's best to check out our "Battery 101" for more information regarding range.

Finish your ride with the Bonfire

1. Unfold the side stand and make sure the Bonfire is stable.
2. To turn off your motorcycle, turn the key to the left and remove it.
3. We recommend always removing the batteries to protect them from theft. To remove the batteries, press lightly against the side door of the battery box and turn the lock to the right.
4. Now slide the fuse up, disconnect the connection to the vehicle and pull it out carefully
5. Complete your Bonfire, see chapter 5.3

During the ride*Brakes*

Please note that unlike a normal gasoline motorcycle, both the rear and front brakes are on the handlebars and there is no foot brake. The combined rear and front brakes are on the left side of the handlebars and the front brake is on the right side of the handlebars.

Driving mode switch

You can change the driving mode with the 1,2,3 key. There are 3 different performance modes to choose from when driving. Never ride beyond your comfort level and always ride according to your personal riding abilities. We always recommend new riders start with the first mode to get used to the bike and the differences between an electric and internal combustion engine. The 3 different power modes have different characteristics in terms of top speed, acceleration and torque to suit riders of all levels, from beginners to professionals.

Power mode 1 - Eco

This mode is always recommended when you ride a Bonfire for the first time. It has a smooth acceleration.

Power mode 2 - Normal

With a top speed of 65 km/h and good acceleration, this mode is perfect for city driving.

Performance mode 3 - Sport

There are no power restrictions here. This mode is perfect when you need all the power available. 80 km/h top speed and good acceleration.

Regenerative braking

When the throttle is released, your bike starts regenerative braking. This allows the bike to stop without using the brakes and recharges the battery.

Important: Regenerative braking can be unreliable, especially when the battery is fully charged. Therefore, always use the regenerative braking function with caution.

Good to know*Side stand*

The side stand can be conveniently folded out even when you are sitting on your bike and activates a switch that cuts the power to the motor for safety reasons. As long as the side stand is folded out, the bike does not move.

Mirror

You can easily adjust the alignment of the mirrors. The joint works with a counter-rotating screw system. Place the mirrors in the thread on the handlebars & turn them clockwise. Align the mirrors and then tighten the nut.

Display

On the display you can read the SOC (state of charge), your riding mode, speed, odometer and time.

Important: It's best to check out our "Battery 101" for more information regarding reading the display.

Operation of the display

The lower left key is the selection key and the upper left key is the setting key.

Clear odometer

With the display on, simply hold down the settings button for 3 seconds. This will clear the trip meter.

Engine control lamp

The engine control light (orange LED next to the ignition lock) should light up for 2 seconds when the vehicle is started and then go out again if there are no faults in the electrical drive train. Please note that after a fault has occurred, the engine control light will remain on for up to three further trips and only then go out again.

The following problems will cause the check engine light to illuminate: undervoltage, overheating engine, overheating controller, or electrical throttle fault.

Emergency operation

You can recognize emergency operation by the fact that the engine control lamp lights up and the vehicle drives significantly slower, i.e. with reduced torque.

Battery undervoltage

If the voltage is outside the normal range of the battery, e.g. in the case of an empty battery, the power is limited in order not to damage the battery or other components of the drive train.

Overheating of the controller

If the controller gets too hot, performance is gradually reduced. If the temperature reaches 100 °C, the bike will stop running until the temperature drops below 80 °C. It is best to take a short break to allow the controller to cool down.

Engine overheating

If the engine gets too hot, the power will be gradually reduced. When the temperature reaches 150°C, the motorcycle will stop running until the temperature falls below 130°C. It is best to take a short break to allow the engine to cool down.

Battery overtemperature and battery undertemperature

The battery does not tolerate extremely warm or cold temperatures. To avoid damage to the battery, it is recommended to leave the bike inside when the outside temperature is below 2°C or above 35°C.

3. Regular inspection and maintenance

3.1. Before and after each trip

To ensure that your bike is always in good condition and to avoid injury or damage, please make sure that you always maintain your Bonfire with the following steps.

- Check your entire motorcycle (according to chapter 3.2.)
- Check the wheels (see chapter 3.3.)
- Check if the brakes are working properly (see chapter 3.4.)
- Clean your motorcycle regularly (see chapter 5.2 - Cleaning)

3.2. Maintenance table and maintenance intervals

Maintenance	Each ride	10 Hours	20 Hours	6 Months
Check the angle of your mirrors	x			
Check that the lights and turn signals are working properly.	x			
Check if the horn is working properly.	x			
Check that the tail light and brake light are working.	x			
Check the torque of fork, axle, handlebar and suspension	x			
Check tire pressure	x			
Carefully check the tires for damage		x		
Check rims for damage		x		
Check spoke tension		x		
Check brake discs		x		
Check brake pads			x	
Check brake fluids				x
Check brake hoses for damage				x
Check front wheel bearing				x
Check the screws on the brake discs			x	
Check footrests			x	
Check battery box				x
Check frame and swing				x
Check wiring harness				x
Check battery connections				x

3.3. Wheels and tires

The wheels should be inflated to a pressure of 230 kPa.

Check your spokes regularly. You must make sure that there are no loose spokes in the wheel.

- Check that the tires are inflated to the correct pressure.
- Check that the tire tread is at least 3 mm deep.
- Check for bent spokes
- Check that all spokes have the correct tension.
- Check that the brake discs are straight.
- Check that the front wheel axle is properly tightened.
- Check that the rear axle is properly tightened.
- Check if the axle of the rear wheel is adjusted correctly.

3.4. Brakes

Inspect the front and rear brake calipers to make sure they or the brake mounts are not damaged. Check the front and rear brake pads to make sure they are not worn. Look through the window into the brake fluid reservoir to see if there is enough brake fluid left. Do a brake test to make sure the front and rear brakes are working properly.

4. Removing and charging the r battery

4.1. Removing the battery

For safety reasons, you cannot ride the bike while charging. Make sure the bike is secure before removing the battery. Use your key to open the battery box. If your bike is on a level surface, the door should remain open. Please turn off the battery. You will find the on/off switch on top of the battery. Now disconnect the battery from the bike. Slide the fuse of the battery upwards. Use the handle of the battery to take it out of the bike.

Important: Always be careful when handling the battery. If it is dropped, visually inspect it and contact a Black Tea Motorbikes representative if it is damaged.

4.2. Charging de r Battery

Plug the charger into the wall socket and then connect the battery. You should be able to hear the charger while it is charging. The charger light will change from green to red when the battery is connected. When the battery is finished charging, the light will turn green.

Important: It's best to check out our "Battery 101" for more information.

4.3. Reinstalling the battery

Make sure that the battery switch is in the "OFF" position. Now put the battery back into the bike. Connect it. Slide the fuse down and turn the battery on. Close and lock the battery box. Don't worry if you ever forget to disconnect the battery and the connectors spark on contact. That's okay, but can wear out the connector over time.

Important: Make sure that the plugs are connected correctly, otherwise the cables and the plugs can be damaged.

5. Ownership and operation

5.1. Maintenance and Cleaning

It is important to keep your Bonfire in good condition to prolong its life. Clean the bike after riding and follow the "Regular Inspection" checklist. Charge the bike to a state of charge of at least 30%. Turn off the bike and the battery if it is not used for more than a week. Otherwise, it will be in standby mode and discharge the battery to an unhealthy level, shortening battery life.

You must always be careful when cleaning the Bonfire. The electrical components of the bike can be damaged if not cleaned properly.

Make sure the motorcycle is stationary and turned off before washing.

Never use a pressure washer or a water hose to clean the motorcycle. It is recommended to use compressed air and a bucket of water together with a cloth and a brush.

5.2. Storage

Follow these steps to ensure that your Bonfire is always in good condition and to extend the life of the motorcycle. We recommend that you ride the bike at least once a month, as some components of the bike need to be used to keep them in the best condition. Store the bike in a closed space and protect it from sunlight to avoid unnecessary aging or discoloration. Make sure your Bonfire is completely turned off by disconnecting the battery.

Never store your Bonfire with an empty battery. If the bike is stored for an extended period of time, charge it to at least 50% to extend the life of the battery.

Never store your Bonfire below 10° C. If you cannot store the motorcycle at higher temperatures, remove the battery and bring it to your home.

Important: It's best to check out our "Battery 101" for more information regarding battery storage

5.3. Interlock

It is recommended to use an additional external lock, which is recommended for motorcycles. When you park your Bonfire on the road, always turn it off, remove the batteries, lock your handlebars and take your key with you.

The steering lock is located under the triple clamp between the forks. Lock the steering lock by turning the handlebars all the way to the left, inserting the key into the steering lock on the right side and turning the key. You will feel the handlebars lock in this position, preventing you from riding your Bonfire.

5.4. Accident

In the event of an accident, please inspect your Bonfire for visual or structural damage. If you notice damage that affects the handling of the vehicle, stop driving to avoid further damage to the vehicle, yourself, or the surrounding area.

If your Bonfire needs repair, please contact us at service@blackteamotorbikes.com.

To check, please go through the procedures in Chapter 3 "Periodic Inspection and Maintenance". In case of a broken or damaged part, visit www.blackteamotorbikes.com to order replacement parts. If you cannot find what you need or have any questions, please contact Black Tea Motorbikes directly.

5.5. Troubleshooting

If your motorcycle is not working properly, please read through the following use cases to find a solution. If you can't find your problem or a working solution, contact Black Tea Motorbikes directly.

My motorcycle does not turn on

First check if the battery, ignition lock and kill switch are properly turned on. Open the battery box to check that the battery is properly connected to the motorcycle. Make sure that the side stand is up.

If your motorcycle still does not turn on, your ignition switch may be defective. Please contact our customer service.

My motorcycle does not give full power

The motorcycle protects itself and may limit performance under certain circumstances. This happens when one of the components of the drivetrain is overheated, under- or over-tensioned, etc.

First, make sure the motorcycle is properly charged by checking the gauge. When the battery is almost empty, the battery cells inside lose the ability to achieve a high discharge (full power).

Please be careful when checking if the controller or the engine are overheated, as you can easily get burned. Please allow the vehicle to cool down sufficiently to avoid permanent damage to the engine, controller or battery.

My engine makes strange noises or does not turn over

If the engine makes strange noises when you step on the gas and does not run properly, please turn off the motorcycle and do not step on the gas, as the engine can still overheat from the battery current. This could happen if the sensors in the engine get wet. The problem is temporary, please put the motorcycle indoors for 24 hours to dry.

My motorcycle does not turn on after a long period of inactivity

Turn on the main switch of the battery and measure the voltage with a multimeter. If the voltage is below 43 V, please charge your battery.

5.6. Tools

To properly maintain your Bonfire and keep it in good condition, we recommend that you always use quality tools and treat the motorcycle with care. If you are unable or unwilling to perform maintenance yourself, please contact someone who has the appropriate experience.

For the Bonfire, Black Tea Motorbikes recommends the purchase and use of the following tools.

- Hexagon wrench set, 1-10 mm
- Screwdriver
- Rubber mallet
- Wrench set 8 - 24 mm
- Ammeter
- Tire iron
- Spoke wrench
- Bleeding kit for brake fluids

6. General vehicle overview

6.1. General vehicle specifications

Article	Specifications
Drive train	3 kW electric PSM hub motor
Driving modes	Eco/ Normal/ Sport
Range per battery (km)	70 / 60 / 55
Maximum speed (km/h)	80
Battery type	Removable 52 V lithium-ion battery pack
Number of battery slots	2
Anti-theft protection	Steering wheel lock
Front fork/suspension	Heavy duty aluminum fork
Rear suspension	Two adjustable 335 mm shock absorbers
Basic dimensions (L x W x H)	1900 mm x 700 mm x 1250 mm
Wheelbase	1300 mm
Permissible total weight	260 kg
Total weight (with 1 battery)	97,5 kg
Total dry weight (without battery)	86 kg

6.2. Electronics specifications

Article	Specifications
Throttle grip	DOMINO Hall (IP66 waterproof)
Battery voltage range	44 V - 58.8 V
Battery dimensions	310 mm x 220 mm x 100 mm
Battery weight	11,5 kg
Battery capacity	1.8 kWh
Rated current	75 A
Peak current	~90 A
Charge current	10 A
Battery charger output	10 Amps @ 58.8 VDC
Charging time	80% charge = 3 hours
Battery cell type	3.7V Lithium-ion - NCM 18650 cells

Article	Specifications
Configuration of the battery pack	14S12P
Battery Management System (BMS)	Safe and efficient charging. Balancing of each cell. Overcharge protection (cut-off at 4.25 V +/- 50 mV). Protection against overdischarge
Battery connection	QS8-S connector with resistors
Battery life cycle	Designed for 800 charging cycles
Motor controller type	Electronic sine wave controller (BLDC)
Controller features	Enhanced regenerative braking; hardware over-current and over-voltage protection; LED fault indication; heat and current reduction and battery undervoltage protection shutdown; aluminum housing for maximum heat dissipation;
Controller environmental protection	IP 66
Motor type	Brushless permanent magnet (50H) 3000W hub motor
Maximum torque	195 Nm
Rated power	3 kW at 450 rpm
Peak performance	5.4 kW at 480 rpm
Engine environmental protection	IP 54
Motor temperature protection	Thermal probe (sensor to prevent overheating)
Wiring	Fully insulated and wrapped wiring harness with waterproof connectors for motor vehicles

6.3. Brakes Specifications

Article	Specifications
Brake system type	CBS hydraulic disc brake
Front wheel brake disc	265 mm disc brake with 3 mm thickness
Rear brake disc	180 mm disc brake with 3 mm thickness
Brake fluid type	DOT-4
Front brake caliper style	Three-piston
Front Brake Pad Type Material	Asbestos-free
Brake caliper rear style	Double piston
Rear Brake Pad Type Material	Asbestos-free

6.4. Wheels and tires specifications

Article	Specifications	
Rim size (front)	1.85 x 18 "	
Rim size (rear)	2.15 x 18 "	
Tire configuration	Spoked rim + rim tape + tube + tire	
Tire options	Street	Offroad
Tire manufacturer	Heidenau	Heidenau
Tire model	K60	K36
Tire size (rear)	110/80 -18 "	110/80 -18 "
Tire size (front)	90/90 - 18 "	3.50 - 18 "
Load capacity of the tire	52 (200 kg)	52 (200 kg)
Tire speed class	S (180 km/h)	P (150 km/h)
DOT safety code for tires	DOT N6YX CB2R	DOT N6YX CB2R

7. Contact and support

If you have any problems or questions about Bonfire, email us at service@blackteamotorbikes.com.

For general questions and information, you can contact us at hi@blackteamotorbikes.com.

Black Tea Motorbikes Ltd.
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This manual is updated regularly.

Please make sure you have the latest version at www.blackteamotorbikes.com/service.



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